Exploring Control Strategies for Drug Delivery: Toward an MPC Approach in an Open-Source Patient Simulator

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***Abstract*—This work presents the development and comparative analysis of control strategies for multi-drug infusion in an open-source physiological patient simulator. The simulator models the pharmacodynamic effects of five anesthetic and hemodynamic drugs using a combination of linear state-space systems and nonlinear functions that capture drug-specific effects and inter-drug interactions. Several approaches are explored, beginning with open-loop offline optimizations for individual drug-target pairs, combined with a feedforward Proportional-Integral (PI) closed-loop regulation. However, strong cross-interactions between drug effects limit the performance of these strategies, which motivates the adoption of a Model Predictive Control (MPC) approach to explicitly handle multi-variable coupling. The MPC formulation is designed by carefully selecting the weights in the cost function to ensure effective control performance, and by introducing nonlinear constraints on specific outputs to reflect physiological safety limits. Additionally, a study of the steady-state gain matrix is conducted to better understand the physiological interactions between drugs and their collective impact on control performance. Simulation results demonstrate that the proposed MPC strategy achieves effective regulation of targeted physiological variables while adhering to drug-specific constraints and minimizing control effort. This work lays the groundwork for more realistic scenarios, including state estimation and embedded implementation for clinical decision support systems.**

I. Introduction

Computer-controlled drug delivery systems are gaining increasing attention in anesthesia, where the administration of multiple interacting drugs must be carefully coordinated to maintain patient safety and optimal physiological conditions for surgical intervention. Traditionally, clinicians rely on manual titration based on experience and observed physiological responses. However, this approach can be suboptimal in complex scenarios involving multiple drug effects and interdependencies, as it is inherently subjective and prone to inter-operator variability. Decisions may be slower, less consistent, and potentially less optimal, especially under high-stress conditions or in rapidly evolving patient states.

Several control strategies have been proposed in the literature for single-drug and multi-drug anesthesia management, including proportional-integral-derivative (PID) control, adaptive control, and more recently, model predictive control (MPC). For instance, a PID-based architecture combined with an optimized feedforward bolus was proposed for propofol administration during the induction phase, achieving faster time-to-target compared to a purely feedback approach [1]. Other studies have explored the closed-loop control of neuromuscular blockade using PID controllers, analyzing the system's dynamic behavior through bifurcation analysis and highlighting risks of oscillations or instability due to nonlinearities [2]. More recently, MPC has gained attention for its ability to explicitly handle multivariable interactions, constraints, and prediction over time horizons. An MPC-based approach was proposed and validated using a multi-drug simulator to control both anesthetic and hemodynamic variables, demonstrating robustness to subsystem coupling and variability [3]. Yet, the development and validation of such strategies often rely on simplified models or proprietary simulators, limiting reproducibility and accessibility.

This project investigates the design and comparative evaluation of control strategies for multi-drug infusion using an open-source patient simulator developed by Clara M. Ionescu and her colleagues at Ghent University [4]. The simulator models the pharmacodynamic effects of five commonly used anesthetic and hemodynamic drugs (propofol, remifentanil, dopamine, nitroprusside and atracurium) on five physiological targets (hypnosis, analgesia, neuromuscular blockade (NMB), mean arterial pressure (MAP), and cardiac output (CO)). It combines linear state-space dynamics with nonlinear pharmacological interactions, offering a realistic testbed for control design.

The goal of this work is to evaluate control strategies for multi-drug infusion, starting with a combined open-loop optimization and feedforward-PI feedback control approach. Due to the limited performance observed in the presence of strong inter-drug couplings and clinical constraints, the study then transitions toward a multi-variable MPC strategy, designed to explicitly address these challenges. By systematically building up from simpler methods and analyzing their limitations, this study highlights the practical challenges of multi-drug control and the benefits of MPC in such contexts.

The rest of this paper is organized as follows. The second section begins by describing the patient simulator and the model reconstruction process, followed by the selection of reference targets. Then, two control strategies are introduced: a preliminary feedforward-PI controller and an MPC approach. The MPC subsection includes an interaction analysis to assess drug coupling effects, as well as the detailed control design. The third section presents the main results. The fourth section discusses the limitations of the proposed strategy and possible perspectives. The main conclusions are drawn in section V.

II. Methods

A. Simulator and Model Reconstruction

The patient simulator used in this project, developed by Ionescu et al. [4], is designed to simulate pharmacokinetics—how the body absorbs, distributes, and eliminates a drug—and pharmacodynamics—how the drug produces its therapeutic effects (PK-PD models). The simulator focuses on representing how drugs move through different compartments, such as the bloodstream, muscle tissue, and fat. The system uses a multi-compartmental model, which simulates the distribution and elimination processes of a drug following intravenous administration. Drug effects on the body are often nonlinear and described using Hill functions, which capture the sigmoidal relationship between drug concentration and response ([4]). Additionally, the simulator accounts for synergetic or antagonistic drug interactions,when drugs are co-administered. These interactions are represented through nonlinear functions that describe how drugs influence each other's effects when administered together. In a synergistic interaction, two drugs might enhance each other's effects, while in an antagonistic interaction, one drug might reduce the effectiveness of the other. These interactions are incorporated into the model to reflect the complex pharmacological mechanisms of drug combination therapies.

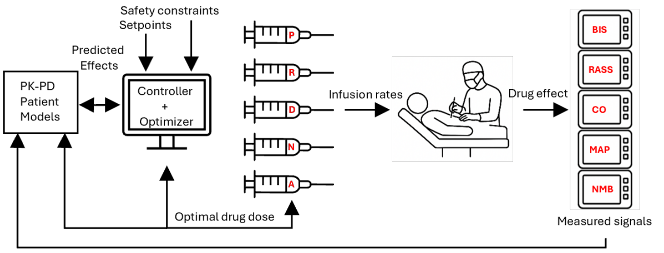

Fig. 1: Schematic representation of the anesthetic-hemodynamic control loop. Adapted from [4].

As shown in Fig. 1, the simulator was developed with the intent of being integrated into a broader closed-loop control system, where optimal drug infusion strategies can be computed in real time. Its purpose is to serve as the core dynamic model within a control framework that aims to regulate the patient's physiological responses during surgery by adjusting the administration of multiple drugs.

The simulator is implemented as an open-source Matlab/Simulink model. As stated by the authors, the objective is *"to encourage the community to work in a systematic and fair-to-compare context while developing computer-based control of multi-drug anesthesia regulatory problems"* [4].

It includes five drug inputs—propofol, remifentanil, dopamine, nitroprusside, and atracurium—each affecting specific physiological responses. Correspondingly, the model outputs five key indicators:

- BIS (Bispectral Index): a percentage from 100 (awake) to 0 (deep anesthesia), reflecting the hypnotic effect of anesthetics;
- RASS (Ramsay Agitation and Sedation Scale): a clinical score ranging from +4 (combative) to -5 (unarousable), used to quantify patient sedation levels;
- CO (Cardiac Output): measured in liters per minute ($\mathrm{L\,min^{-1}}$), indicating heart performance;
- MAP (Mean Arterial Pressure): in millimeters of mercury (mmHg), reflecting overall blood pressure control;
- NMB (Neuromuscular Blockade): expressed as a percentage from 0% (complete blockade) to 100% (no blockade).

To enable implementation and manipulation of the patient model outside the Simulink environment, the entire simulator was reconstructed in a fully script-based MATLAB form specifically for this project. The modeling approach follows a block-oriented Wiener structure, commonly used to represent systems with nonlinear behavior. In this configuration, the system is described as a linear dynamic block followed by a static nonlinear function, which distinguishes it from a Hammerstein model where the order is reversed [5]. The linear block captures the time-dependent dynamics of the system, while the static nonlinearity represents how the system processes outputs in a nonlinear manner.

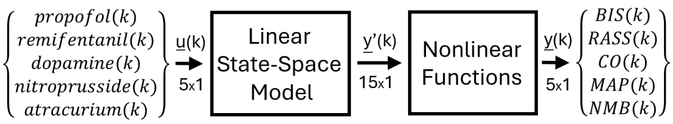

Fig. 2: Block diagram representation of the simulator.

In this case, all the linear dynamic elements were grouped into one large discrete-time state-space model, while all static nonlinearities—primarily Hill functions and interaction terms between co-administered drugs—were gathered into a single nonlinear function. The result is a model structure where the five drug infusion rates are inputs to a global linear state-space system, which produces 15 intermediate outputs. These outputs are then passed through the nonlinear function block, yielding the five final physiological indicators (BIS, RASS, CO, MAP, NMB). The full system operates in discrete time, with a sampling time of 5s. This choice was made to ensure that measurements would not need to be taken too frequently and that input modifications could occur at reasonable intervals. Furthermore, the system demonstrates satisfactory behavior under this sampling rate.

Concretely, the original Simulink model contains 15 state-space models representing linear dynamics, some provided directly as state-space systems and others as transfer functions, which were converted accordingly. All 15 systems were merged into a single global state-space representation by concatenating their individual matrices A, B, C, and D along the diagonal. Although this does not result in a minimal realization, it ensures ease of translation and traceability during implementation. The final global state-space system has dimensions $A \in \mathbb{R}^{74\times 74}$, $B \in \mathbb{R}^{74\times 5}$, $C \in \mathbb{R}^{15\times 74}$, and $D \in \mathbb{R}^{15\times 5}$, and is governed by the standard discrete-time equations:

$$\underline{x}(k+1) = A\underline{x}(k) + B\underline{u}(k), \quad \underline{y}'(k) = C\underline{x}(k) + D\underline{u}(k) \tag{1}$$

where $\underline{u}(k) \in \mathbb{R}^5$ is the vector of drug infusion inputs, and $\underline{y}'(k) \in \mathbb{R}^{15}$ contains intermediate physiological variables.

The final nonlinear function, implemented as a MATLAB function, takes the 15 intermediate outputs and applies all static nonlinearities—including Hill-based dose-response curves and nonlinear drug interaction terms—to compute the five final outputs.

$$\underline{y}(k) = F(\underline{y}'(k)) \tag{2}$$

where $\underline{y}(k) \in \mathbb{R}^5$ is the vector of final outputs (the five indexes), and F is the nonlinear function that takes the intermediate $\underline{y}'(k)$ and computes all the interactions. To validate the reconstruction, both the original Simulink simulator and the MATLAB-based model were subjected to identical input profiles, and their outputs were compared. The results were found to be identical, confirming the accuracy of the reconstruction.

B. Reference Targets

This section details the reference targets for the monitored parameters. Each index has a defined target range, based on its physiological implications and clinical recommendations.

The BIS index, which reflects the level of cortical activity, is typically maintained between 40 and 60% during general anesthesia. This range is associated with a low probability of intraoperative awareness while avoiding excessive anesthesia depth, which could prolong recovery or lead to hemodynamic instability [6]. BIS values below 40% suggest deep hypnosis with potential adverse effects, while values above 60% may reflect inadequate anesthesia. In this work, a value of 50% was chosen for the BIS target.

The RASS is used to assess the level of sedation. During surgery, values between -5 (unarousable) and -2 (light sedation) are generally targeted, depending on the anesthetic technique and patient needs [7]. Deep sedation is necessary to suppress consciousness and discomfort, especially in combination with neuromuscular blockade, but excessive suppression can complicate postoperative recovery. In this work, a target value of -4 was selected.

CO reflects global perfusion and is a critical determinant of tissue oxygenation. A normal CO during anesthesia is typically in the range of $5\,\mathrm{L\,min^{-1}}$ to $7\,\mathrm{L\,min^{-1}}$ in adult patients. Values below this threshold may lead to organ hypoperfusion and increased postoperative morbidity [8]. Therefore, maintaining CO within this range helps ensure sufficient oxygen delivery.

MAP is a surrogate for organ perfusion pressure. A minimum threshold of $65\,\mathrm{mmHg}$ is widely accepted, as it is necessary to maintain perfusion to vital organs such as the kidneys and brain. However, for patients with chronic hypertension or during high-risk surgeries, higher targets (e.g., $80\,\mathrm{mmHg}$) are often recommended to prevent ischemic complications [9].

Lastly, the degree of NMB is monitored to ensure optimal surgical relaxation. Sufficient blockade is typically defined by a train-of-four (TOF) ratio where the first twitch response is between 10% and 20% of baseline. This level minimizes patient movement and facilitates surgical exposure while limiting the risk of residual paralysis at the end of the procedure [2]. In this work, a target value of 15% was chosen, as it lies within the recommended range.

C. Feedforward-PI Control Strategy with Open-Loop Optimization

The control strategy implemented in this project builds upon the method proposed by Schiavo et al. [1], who designed a feedforward-PID controller to regulate the depth of anesthesia based on the BIS. Their approach combines an offline open-loop optimization with a closed-loop feedforward-PID controller. The output of the nonlinear open-loop optimization is the infusion profile used as the feedforward input in the control loop. In the present work, this strategy was adapted by replacing the PID controller with a PI controller, while preserving the feedforward-feedback structure.

To reduce complexity, the system was first simplified to a single-input single-output (SISO) configuration, considering only the effect of propofol infusion (input) on the BIS index (output). An optimal infusion profile was computed offline using MATLAB's `fmincon` solver by solving the following constrained optimization problem.

The objective was to:

- Minimize the time k_{opt} at which the BIS first reaches the target level $Y_0 = 50$.
- Ensure the BIS remains close to 50 after k_{opt}.
- Avoid excessive drug administration.

These objectives were encoded in the following cost function:

$$J = k_{\text{opt}} + \alpha \cdot \frac{1}{N} \sum_{k=k_{\text{opt}}}^{N} (y_k - 50)^2 + \beta \cdot \sum u_{\text{prop}}(k) \tag{3}$$

where y_k is the simulated BIS at time step k, $u_{\text{prop}}(k)$ is the propofol infusion rate, and $\alpha = \beta = 0.01$ are weighting factors.

The optimization was subject to the following constraints:

$$y_k \geq 40, \qquad \forall k \tag{4}$$

$$y_k \leq 50, \qquad \forall k \geq k_{\text{opt}} \tag{5}$$

$$0 \leq u_{\text{prop}}(k) \leq 6.67[\text{mg/s}], \qquad \forall k \tag{6}$$

The optimal propofol infusion profile and the resulting BIS trajectory in open-loop are shown in Figure 3. The left subplot displays the infusion profile, while the right subplot shows the BIS response.

In clinical settings, patient-specific variability and modeling errors may lead to significant deviations from the expected BIS response. To ensure robustness, the open-loop strategy was complemented with a closed-loop feedforward-PI controller that compensates for model mismatches and external disturbances by dynamically adjusting the infusion rate based on the tracking error, thereby improving the overall safety and effectiveness of the drug administration.

The reference trajectory $r(t)$ used by the controller corresponds to the BIS signal obtained from the open-loop simulation using the optimal infusion profile (Figure 3b). The control diagram shown in Figure 4 consists of a feedback loop with a PI controller and an additional feedforward component

(a) (b)

Fig. 3: Optimal infusion profile (left) and resulting BIS trajectory (right) for feedforward control.

$u_{\text{ff}}(t)$. The reference signal $r(t)$ is compared to the system output $y(t)$ to compute the control error, which is processed by the PI controller to generate $u_{\text{PI}}(t)$. This signal is then summed with the feedforward input $u_{\text{ff}}(t)$ to form the total control input $u(t)$. Before being applied to the patient model, $u(t)$ passes through a saturation block representing the physical limitation of the pump.

The total infusion rate is computed as:

$$u(t) = u_{\text{ff}}(t) + u_{\text{PI}}(t) \tag{7}$$

where $u_{\text{ff}}(t)$ is the open-loop optimal profile and $u_{\text{PI}}(t)$ is the output of the PI controller based on the tracking error.

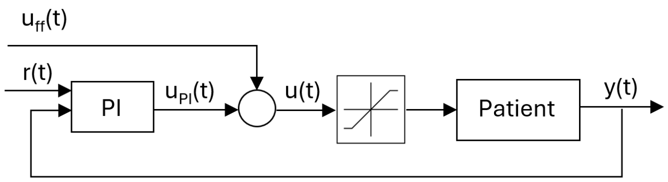

Fig. 4: Block diagram of the feedforward-PI controller.

The PI block is described in the Laplace domain by:

$$U_{\text{PI}}(s) = K_p \left(E(s) + \frac{E(s)}{sT_i} \right) \cdot \frac{1}{(T_f s + 1)^2} \tag{8}$$

with:

- $K_p = 0.16\,\text{mg}\,\text{s}^{-1}$ (proportional gain),
- $T_i = 476\,\text{s}$ (integral time constant),
- $T_f = 0.7\,\text{s}$ (output filter time constant).

As shown in Figure 5, the controller successfully maintained the BIS close to its reference in the SISO configuration without drug interaction.

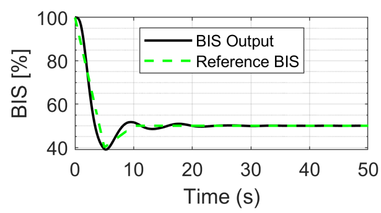

Fig. 5: Closed-loop BIS regulation using the feedforward-PI controller in a SISO configuration.

The same control strategy was applied to regulate the RASS index using remifentanil infusion. The feedforward-PI controller successfully tracked the reference trajectory in this SISO context (Figure 6).

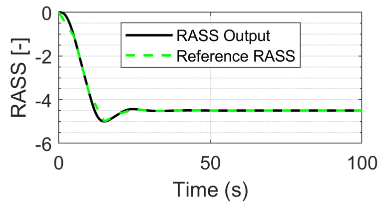

Fig. 6: Closed-loop RASS regulation using a feedforward-PI controller in a SISO configuration.

However, when both drug-control loops were combined to form a multi-input multi-output (MIMO) system, the strong pharmacodynamic influence of remifentanil on the BIS index significantly impaired the ability of the independent feedforward-PI controllers to regulate BIS. As a result, the BIS could no longer be properly controlled and deviated substantially from its target (Figure 7). In contrast, the RASS output remained close to its reference, as it was not affected by the propofol infusion. This performance imbalance highlighted the limitations of decentralized control in the presence of drug interactions and motivated the transition to an MPC strategy.

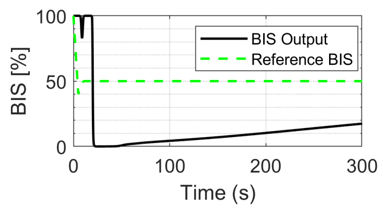

Fig. 7: Failure of BIS regulation under decentralized feedforward-PI control in a MIMO configuration.

D. Model Predictive Control

1) Interaction Analysis: Prior to MPC implementation, a fundamental analysis of the interactions between inputs and outputs was conducted. Understanding which drug affects which clinical output allows for better tuning of the control strategy. For this purpose, a steady-state gain analysis was conducted. A local approximation of the steady-state gain matrix $G \in \mathbb{R}^{5\times5}$ was computed using the product of the following two steady-state gain components:

- $G_{\text{linear}} \in \mathbb{R}^{15\times5}$: the gain of the linear state-space model, mapping drug infusion rates to intermediate variables $\underline{y}'$.

- $G_{\text{nonlinear}} \in \mathbb{R}^{5\times 15}$: the Jacobian of the nonlinear function mapping $\underline{y}'$ to the final clinical outputs $\underline{y}$.

The matrix G_{linear} was obtained by applying successive unit step inputs to each drug input channel u_i, with all other inputs held at zero. For each input, the steady-state internal output $\underline{y}'$ was recorded, yielding:

$$G_{\text{linear}}[:,i] = \lim_{t\to\infty} \underline{y}'(t) \quad \text{with } u_i(t) = 1,\ u_{j\neq i}(t) = 0 \qquad (9)$$

This procedure resulted in a matrix of size 15×5, consistent with the input dimension 5×1 and intermediate output dimension 15×1.

The matrix $G_{\text{nonlinear}}$ (dimension 5×15) was derived from the Jacobian of the nonlinear output mapping function F, evaluated at a nominal operating point:

$$G_{\text{nonlinear}} = \left.\frac{\partial F}{\partial \underline{y}'}\right|_{\underline{y}'=\underline{y}'_{\text{eq}}} \qquad (10)$$

The chosen equilibrium point $\underline{y}'_{\text{eq}}$ corresponds to the steady-state response of the system under the drug input vector:

$$u_{\text{eq}} = \begin{bmatrix}0.3 & 1.8 & 0 & 0 & 6.5\end{bmatrix}^\top \qquad (11)$$

The final local steady-state gain matrix $G \in \mathbb{R}^{5\times 5}$ was then computed as the product:

$$G = G_{\text{nonlinear}} \cdot G_{\text{linear}} \qquad (12)$$

The final steady-state input-output relationship can be expressed as:

$$\begin{bmatrix}y_1\\y_2\\y_3\\y_4\\y_5\end{bmatrix} \approx \begin{bmatrix}-208.03 & -47.08 & 60 & 0 & 0\\ 0 & -2.32 & 0 & 0 & 0\\ 0 & 0.76 & 4.96 & 12 & 0\\ -0.01 & -18.45 & 3 & -15 & 0\\ 0 & -0.55 & 0 & 0 & -4.8\end{bmatrix} \cdot \begin{bmatrix}u_1\\u_2\\u_3\\u_4\\u_5\end{bmatrix} \qquad (13)$$

with y_1 = BIS, y_2 = RASS, y_3 = CO, y_4 = MAP, y_5 = NMB and u_1 = propofol, u_2 = remifentanil, u_3 = dopamine, u_4 = nitroprusside, u_5 = atracurium. To enable a fair comparison of the relative influence of each drug on the clinical outputs, the gain matrix should be normalized according to the typical variation ranges of the indices; BIS typically varies from 100 to 0%, RASS from 5 to -5, CO from 0 to 10 $\mathrm{L\,min^{-1}}$, MAP from 80 to 60mmHg, and NMB from 100 to 0%. Accordingly, the rows of the gain matrix were normalized by dividing them respectively by 100, 10, 10, 20, and 100.

$$G_{\text{nonlinear}} \approx \begin{bmatrix}-2.08 & -0.47 & 0.6 & 0 & 0\\ 0 & -0.23 & 0 & 0 & 0\\ 0 & 0.08 & 0.50 & 1.20 & 0\\ 0 & -0.92 & 0.15 & -0.75 & 0\\ 0 & -0.01 & 0 & 0 & -0.05\end{bmatrix} \qquad (14)$$

From the normalized gain matrix, several key observations can be made regarding drug influence on physiological indices. BIS (y_1) is primarily strongly decreased by propofol (u_1), with a moderate decrease due to remifentanil (u_2) and an increase associated with dopamine (u_3). RASS (y_2) is affected exclusively by remifentanil, which induces a moderate decrease. CO (y_3) is significantly increased by both nitroprusside (u_4) and dopamine, with a minor contribution from remifentanil. Notably, there is no pharmacological action that counteracts the rise in CO, which indicates that its rise will need to be constrained through an upper bound within the MPC framework. MAP (y_4) is strongly reduced by both remifentanil and nitroprusside, and can be increased by dopamine, enabling bidirectional regulation. Finally, the NMB (y_5) is almost exclusively decreased by atracurium (u_5). These observations enable the construction of a comparative influence table, which revises the one originally presented in Ionescu *et al.* [4].

Input / Output	BIS	RASS	CO	MAP	NMB
Propofol	↓↓	–	–	–	–
Remifentanil	↓	↓	↑	↓↓	–
Dopamine	↑	–	↑	↑	–
Nitroprusside	–	–	↑↑	↓↓	–
Atracurium	–	–	–	–	↓

TABLE I: Summary of the main drug influences on physiological outputs.

2) MPC Design: MPC is a modern control strategy that solves an open-loop optimal control problem at each sampling instant, using the current system state as the initial condition [10]. At each time a new measurement becomes available (i.e., at each sampling step), an optimization problem is formulated over a finite prediction horizon, yielding a sequence of future control actions. However, only the first control input of this sequence is applied to the system, and the optimization is repeated at the next measurement—this is known as the moving horizon strategy. Although each optimization is open-loop, the repeated feedback of updated measurements at each step renders the overall control policy closed-loop. A reference model is introduced to generate a desired output trajectory. The control objective is to minimize the deviation between the predicted outputs and this reference trajectory, while also considering the cost associated with the control effort. This leads to a multi-objective optimization formulation, balancing performance tracking and input regulation under possible system constraints.

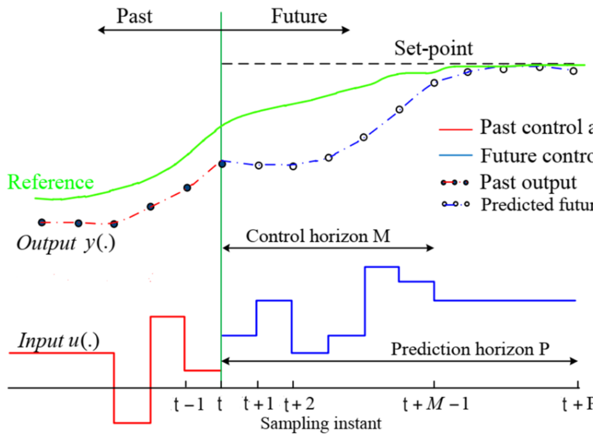

Fig. 8: Illustration of MPC principle from course slides of Optimal Control [10].

The MPC strategy is particularly well-suited for systems with multiple interacting inputs and outputs. As the patient system has five inputs and five outputs with significant interactions, MPC can optimize control actions for each input while accounting for the coupling between variables. The MPC also has the ability to directly incorporate physical constraints on both inputs and outputs. In this project, the MPC controller was designed with fixed reference setpoints for BIS, RASS, and NMB, and range-based constraints for CO and MAP. The setpoints for BIS, RASS, and NMB were respectively fixed at 50%, -4, and 15%. These values were selected based on their critical importance in ensuring optimal surgical conditions, as discussed in Section II-B. In contrast, CO and MAP were not regulated toward fixed references. Instead, admissible ranges were imposed to reflect the physiological tolerance observed in clinical practice. Specifically, the lower bounds for CO and MAP were set to $5\,\mathrm{L\,min^{-1}}$ and $65\,\mathrm{mmHg}$, respectively, while the upper bound for CO was limited to $10\,\mathrm{L\,min^{-1}}$. No upper constraint was imposed on MAP. The objective function minimized by the controller was designed to penalize deviations from the reference values over the prediction horizon. Let $y_i(k)$ denote the predicted value of output i at future time step k, and y_{ri} its reference. The cost function is given by:

$$J = \sum_{k=1}^{N_p} \Big(w_{\mathrm{BIS}} \left(y_{\mathrm{BIS}}(k) - y_{\mathrm{rBIS}} \right)^2 + w_{\mathrm{RASS}} \left(y_{\mathrm{RASS}}(k) - y_{\mathrm{rRASS}} \right)^2 + w_{\mathrm{NMB}} \left(y_{\mathrm{NMB}}(k) - y_{\mathrm{rNMB}} \right)^2 \Big) + R \tag{15}$$

where $w_{\mathrm{BIS}} = 0.02$, $w_{\mathrm{RASS}} = 0.2$, and $w_{\mathrm{NMB}} = 0.012$. The term R is given by:

$$R = 10 \sum_{i=1}^{N_p} \left(u_3(i)^2 + u_4(i)^2 \right) \tag{16}$$

where u_3 and u_4 represent the doses of dopamine and nitroprusside, respectively. The weights $w_{\mathrm{BIS}}, w_{\mathrm{RASS}}, w_{\mathrm{NMB}}$ were selected inversely proportional to the expected variation range of each index. This normalization ensures that each component contributes comparably to the total cost function, despite differing units or scales. For the constraints on CO and MAP, here are the equations:

$$\begin{cases} CO_{\min} < CO < CO_{\max} \\ MAP < MAP_{\min} \end{cases} \tag{17}$$

As already discussed, input constraints must be implemented, ensuring that the administered doses of drugs do not exceed the syringe capacity. The minimum dose for each drug is set to $0\,\mathrm{mg\,s^{-1}}$, and the maximum allowable dose is $6.67\,\mathrm{mg\,s^{-1}}$.

For the prediction horizon, N_p is set to $100\,\mathrm{s}$, and for the control horizon, N_c is set to $50\,\mathrm{s}$. These values were chosen by trial and error to ensure a sufficient prediction of the system behavior while maintaining a manageable control horizon for real-time implementation. The distinction between these two horizons lies in the number of future inputs being optimized. Specifically, only the first N_c control inputs are computed during optimization, forming the matrix M. For the remaining $N_p - N_c$ steps, the last computed input is held constant.

Let $M \in \mathbb{R}^{5 \times N_c}$ be the matrix of optimal drug infusion profiles over the control horizon, where each row corresponds to a drug u_i for $i = 1, \ldots, 5$, and each column corresponds to a time step. To evaluate the cost function over the full prediction horizon N_p, the input profile is extended by holding the control inputs constant beyond N_c, using the last computed column $M(:, N_c)$. The resulting extended input matrix $\tilde{M} \in \mathbb{R}^{5 \times N_p}$ is defined as:

$$\tilde{M} = \left[M \quad \underbrace{M(:, N_c) \quad \ldots \quad M(:, N_c)}_{N_p - N_c \text{ times}} \right] \tag{18}$$

This approach reduces the number of decision variables while still allowing the prediction model to simulate system behavior over the full prediction horizon.

The controller was implemented using the `fmincon` solver at each sampling time to compute the optimal sequence of drug infusions over the control horizon, ensuring that all input and output constraints were satisfied. Only the first control input was applied at each step, in line with the receding horizon principle. The block representation of the MPC control-loop is represented on the Figure 9. This strategy was applied to the system over a 300-second simulation. The resulting output trajectories and corresponding drug infusion profiles are presented and analyzed in the following section.

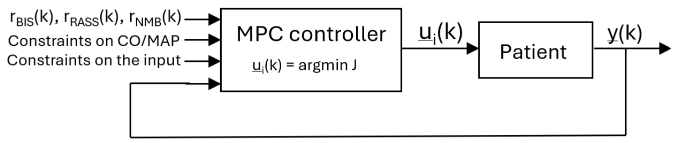

Fig. 9: Block diagram of the MPC controller.

III. Results

The results obtained from the simulations are satisfactory. The fixed setpoints for BIS, RASS, and NMB are accurately tracked (see Figures 10a, 10b, 10c), with each variable converging to its respective target value without deviation. Specifically, the BIS reaches its desired value after 18.7 seconds, the RASS after 23.1 seconds, and the NMB after 80 seconds. These times are clinically acceptable: the patient is brought into a deep unconscious state in less than 20 seconds, and full neuromuscular blockade is achieved in approximately 80 seconds, ensuring optimal conditions for surgical intervention. The system efficiently maintains these setpoints, guaranteeing stability and continuity of anesthesia during the procedure.

For CO and MAP, the imposed constraints are respected throughout the simulation (see Figures 10d, 10e). The CO remains within the admissible range of [5, 10] $\mathrm{L\,min^{-1}}$, while the MAP stays above the lower bound of $65\,\mathrm{mmHg}$, without violating the constraints. Additionally, the total computation time required to solve the online optimization over the 300-second simulation horizon was approximately 200 seconds. This indicates that the controller can compute optimal actions faster than real time, providing enough margin to apply and adjust drug infusions progressively. These results confirm that the MPC-based controller is effective in meeting both setpoint tracking and constraint satisfaction requirements.

In addition to regulating the physiological outputs, the MPC strategy also provides optimal drug infusion profiles for each of the five administered medications (see Figure 11). These profiles are generated in a piecewise-constant (staircase) form and are updated at every sampling instant (with $T_s = 5s$). This discrete control structure aligns with clinical practice, where drug dosages are typically adjusted in steps rather than continuously.

For propofol, remifentanil, dopamine, and nitroprusside, the majority of the dose is administered at the beginning of the procedure, allowing the corresponding physiological outputs (BIS, RASS, CO, and MAP) to rapidly reach their desired levels. After this initial control phase, only low and steady infusion rates are required to maintain the effect.

Conversely, the infusion profile of atracurium exhibits a more sustained delivery throughout the simulation. This behavior aligns with the gain analysis performed earlier: the neuromuscular block (NMB) output shows relatively low sensitivity to the atracurium input, especially considering the required decrease from 100% to 15%. As a result, larger quantities of the drug must be injected over time to achieve and maintain an adequate level of muscle relaxation. Small oscillations can be observed in the infusion profiles. These are likely numerical artifacts caused by the discretization of the optimization problem and the use of the `fmincon` solver, rather than physiologically relevant variations.

IV. Discussion and Limitations

The simulation results demonstrate that the proposed MPC controller is effective in regulating the patient's physiological variables within the desired bounds during the simulated surgical procedure. However, several considerations should be addressed regarding the fidelity of the patient simulator and the potential limitations of the current approach.

The patient simulator used in this study relies on a detailed PK-PD model, which accurately captures the complex interactions between the administered drugs and the body's physiological responses. While this high-fidelity model provides a precise representation, it also adds significant complexity to the control problem. A simpler, reduced-order model could potentially be sufficient for effective control in many clinical scenarios, reducing both computational costs and the complexity of controller design. The current approach is highly dependent on the specific PK-PD models of the drugs used. If different medications are selected, whether from the same class or a completely different group, the model must be updated accordingly.

In the simulation, all five drugs—propofol, remifentanil, dopamine, nitroprusside, and atracurium—are administered simultaneously to regulate their respective physiological targets. However, in practice, not all of these medications may be used in every case. Depending on the specific requirements of the surgery or the patient's condition, some medications may be optional or replaced by alternatives. If fewer drugs are used simultaneously, the overall system would likely experience fewer interactions between drugs, potentially simplifying control.

Additionally, the current approach assumes that all states of the system are perfectly known at all times, which is not the case in practice. In reality, only a limited number of outputs are measurable, and many internal states cannot be directly observed. Therefore, future work must incorporate state estimation techniques, such as a Kalman filter, to estimate these unmeasured states. Given the high dimensionality and nonlinearity of the system, designing a state estimator for this application will be challenging. The use of an Extended Kalman Filter (EKF) may be necessary to accurately estimate the unmeasured states and improve the real-time applicability of the controller.

V. Conclusion

This project focused on the design and implementation of a control system for multi-drug infusion regulation within a patient simulator. Initial strategies based on closed-loop control with a feedforward-PI controller were found to be inadequate for handling the complex interactions between multiple drugs. In contrast, MPC effectively addressed these limitations, enabling accurate and robust regulation of physiological variables while accounting for multivariable dynamics and clinical constraints.

Simulation results confirmed the effectiveness of MPC in achieving precise control over key physiological parameters. As a next step, the integration of a state estimator will be essential to enable real-time implementation of the controller in scenarios where not all system states are directly measurable.

VI. Acknowledgment

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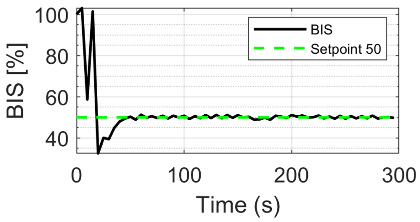
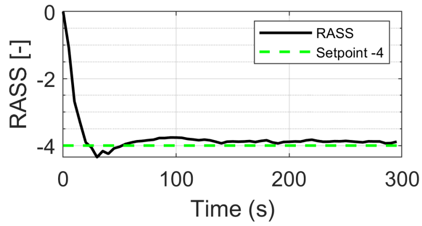
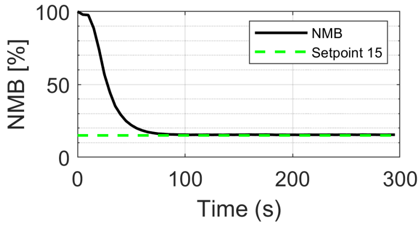


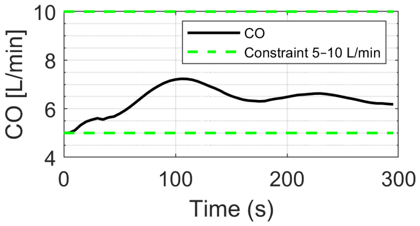
(a) BIS



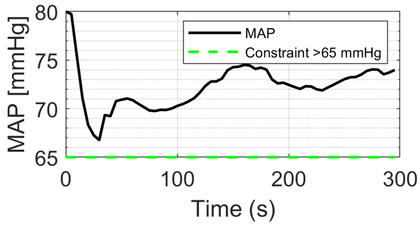
(b) RASS



(c) NMB

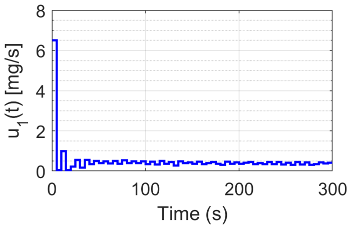


(d) CO

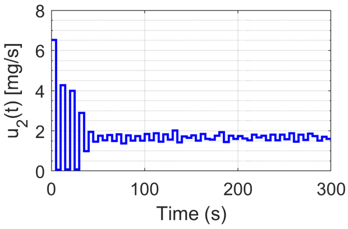


(e) MAP

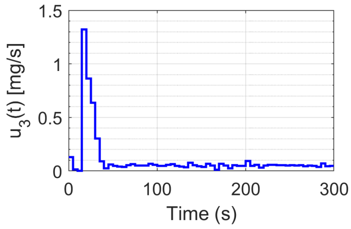
Fig. 10: Time evolution of the BIS, RASS, NMB, CO and MAP outputs under MPC.



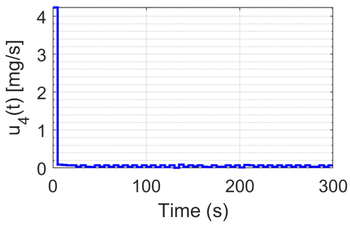
(a) Propofol



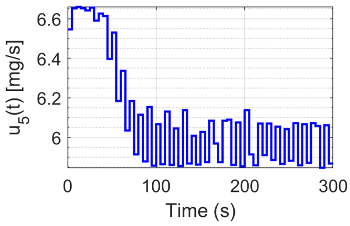
(b) Remifentanil



(c) Dopamine



(d) Nitroprusside



(e) Atracurium

Fig. 11: Optimal infusion profiles for propofol, remifentanil, dopamine, nitroprusside and atracurium computed by the MPC.

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